SQL Code

```sql
CREATE DATABASE employee;
USE employee;
CREATE TABLE departments (
    department_id   INT,
    department_name VARCHAR(100)
);
CREATE TABLE location (
    location_id INT,
    location    VARCHAR(30)
);
CREATE TABLE employees (
    employee_id   INT,
    employee_name VARCHAR(50),
    gender        ENUM('M','F'),
    age           INT,
    hire_date     DATE,
    designation   VARCHAR(100),
    department_id INT,
    location_id   INT,
    salary        DECIMAL(10,2)
);
DESCRIBE departments;
DESCRIBE location;
DESCRIBE employees;
ALTER TABLE employees ADD email VARCHAR(100);

ALTER TABLE employees MODIFY designation VARCHAR(255);

ALTER TABLE employees DROP COLUMN age;

ALTER TABLE employees RENAME COLUMN hire_date TO date_of_joining;

DESCRIBE employees;

RENAME TABLE departments TO departments_info;
RENAME TABLE location TO locations;

SHOW TABLES;

TRUNCATE TABLE employees;


DROP TABLE employees;
DROP DATABASE employee;
DROP DATABASE IF EXISTS employee;
CREATE DATABASE employee;
USE employee;
-- PRIMARY KEY = unique id; NOT NULL + UNIQUE = no null or duplicate names
CREATE TABLE departments (
    department_id   INT PRIMARY KEY,
```

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    department_name VARCHAR(100) NOT NULL UNIQUE
);
CREATE TABLE location (
    location_id INT AUTO_INCREMENT PRIMARY KEY,
    location    VARCHAR(30) NOT NULL UNIQUE
);
CREATE TABLE employees (
    employee_id   INT PRIMARY KEY,                    -- distinct identifier
    employee_name VARCHAR(50) NOT NULL,               -- name always required
    gender        ENUM('M','F'),                      -- only M or F allowed
    age           INT CHECK (age >= 18),              -- must be 18 or above
    hire_date     DATE DEFAULT (CURRENT_DATE),        -- defaults to today
    designation   VARCHAR(100),
    department_id INT,
    location_id   INT,
    salary        DECIMAL(10,2),
    FOREIGN KEY (department_id) REFERENCES departments(department_id),
    FOREIGN KEY (location_id)   REFERENCES location(location_id)
);
DESCRIBE departments;
DESCRIBE location;
DESCRIBE employees;
```